Team Formation

At the beginning of the project, the team conducted an initial meeting to establish the foundation for collaboration and project coordination. The purpose of this meeting was to introduce the team members, discuss their academic backgrounds, technical skills, strengths, and areas of interest. This step helped the team gain a clear understanding of each member's capabilities and ensured that responsibilities could be distributed effectively.

The team members participating in this project are:

- Khalid Mohammed Alomari
- Abdullah Mushabab Alsalem
- Abdulrahman Dhaifallah Alghamdi

During the meeting, the team agreed to assign initial roles to ensure proper coordination during the first stage of the project. A temporary Project Manager role was designated to Abdullah Mushabab Alsalem to organize meetings, monitor progress, and coordinate tasks among team members. These roles are considered flexible and may evolve as the project progresses and new requirements emerge.

In addition, the team established a set of collaboration and communication guidelines to maintain an organized and productive workflow. The team selected Discord as the primary platform for communication, allowing members to share updates, discuss ideas, and coordinate tasks efficiently.

To ensure effective teamwork, the team agreed on the following working principles:

- Encourage active participation from all team members during discussions and decision-making.
- Maintain clear and transparent communication through the agreed communication channels.
- Respect diverse perspectives and promote collaborative problem-solving.
- Ensure accountability by completing assigned tasks within the agreed timeline.

By the end of the team formation stage, the team successfully established a collaborative environment, defined initial coordination roles, and agreed on communication strategies that support efficient teamwork throughout the project.

1. Research and Brainstorming

During the research and brainstorming phase, the team conducted both individual and collaborative research to explore potential project ideas. The goal of this stage was to identify meaningful problems that could be solved through innovative digital solutions.

Team members explored various sources for inspiration, including real-world problems, emerging technology trends, and innovative digital platforms. Special attention was given to solutions that could create real value for users and address practical challenges faced by individuals or organizations.

After the initial research phase, the team organized a structured brainstorming session to generate and analyze potential project ideas. The team applied the "How Might We" questions technique to encourage creative thinking and ensure that multiple perspectives were considered.

"How Might We" Questions

To encourage open-ended thinking, the team used "How Might We" questions to frame potential problems and opportunities. These questions helped guide the discussion toward practical solutions. Examples included:

- How might we support families in managing therapy exercises for children at home?
- How might we provide safer environments for testing cybersecurity defenses?
- How might we simplify the creation of customized dashboards for organizations?
- How might we use artificial intelligence to generate long-form videos while maintaining context and continuity between scenes?

Ideas Generated During Brainstorming

During the brainstorming session, the team explored four potential project ideas:

Idea 1 – SPD Support Platform

A digital platform designed to support children with Sensory Processing Disorder (SPD) by providing a structured system for performing therapeutic sensory exercises at home. The platform allows specialists to create therapy plans and convert them into clear exercises that families can follow. Parents can track the child's progress, record completed activities, and receive guidance on how to perform exercises properly. The system also enables specialists to monitor progress remotely and adjust therapy plans when needed.

Idea 2 – Virtual Penetration Testing Environment

A platform that simulates a secure virtual environment where organizations can safely test the security of their websites or applications. The system would function similarly to a penetration testing laboratory used by red teams to identify vulnerabilities and analyze potential cyber threats.

Idea 3 – Custom Dashboard Builder Platform

A platform that allows organizations to create customized dashboards tailored to their operational needs. The system would provide flexible components and modules that allow businesses to build control panels focused on dashboard customization and data management.

Idea 4 – AI-Based Long Video Generation Tool

A tool that uses artificial intelligence to generate long-form videos while maintaining context from previously generated content. The goal is to enable creators to produce extended video content without losing continuity or narrative coherence.

2. Idea Evaluation

After generating several ideas during the brainstorming phase, the team conducted a structured evaluation process to determine the most suitable idea for the project. The purpose of this stage was to compare the proposed ideas objectively and select the concept that best aligns with the team's skills, project goals, and potential impact.

Evaluation Criteria

The team defined the following criteria to evaluate each idea:

- **Feasibility:** The practicality of implementing the idea within the available time, resources, and technical capabilities.
- **Potential Impact:** The extent to which the solution can benefit users or solve a real-world problem.
- **Technical Alignment:** How well the idea matches the technical skills and experience of the team members.
- **Scalability:** The ability of the solution to grow and support more users or additional features in the future.

Each idea was scored on a scale from 1 (Low) to 5 (High) for each criterion.

Idea	Feasibility	Impact	Technical Alignment	Scalability	Total
SPD Support Platform	4	5	4	4	17
Penetration Testing Lab	3	4	4	3	14
Custom Dashboard Builder	4	3	3	4	14
AI Video Generation Tool	2	4	2	4	12

Based on this evaluation, the SPD Support Platform received the highest score due to its strong social impact and alignment with the team's ability to build a structured digital platform.

Risks and Constraints

SPD Support Platform

- Requires consultation with specialists to ensure exercises are appropriate.
- Needs careful design to make the platform simple for families to use.

Penetration Testing Environment

- High security and infrastructure requirements.
- Potential ethical and legal considerations when simulating attacks.

Custom Dashboard Builder

- Highly competitive market with existing solutions.
- Requires extensive customization features to stand out.

AI Video Generation Tool

- High computational requirements.
- Technical complexity in maintaining context across long video sequences.

Considering these factors, the team concluded that the SPD Support Platform offers the best balance between feasibility, impact, and innovation.

3. Decision and Refinement

After completing the brainstorming and idea evaluation stages, the team held a discussion to review the evaluation results and determine the most suitable concept for the project's Minimum Viable Product (MVP). Based on the evaluation criteria and scoring process, the team agreed to select the SPD Support Platform as the final MVP idea.

The decision was primarily influenced by the idea's strong social impact, clear real-world problem, and its feasibility within the project's timeline and the team's technical capabilities.

Problem the Project Solves

Many families with children diagnosed with Sensory Processing Disorder (SPD) face difficulties in consistently following therapeutic exercises at home. Although specialists design appropriate therapy plans that include various sensory exercises and activities, some parents may struggle to fully understand the exercises, apply them correctly, or consistently track their child's progress.

In addition, specialists in hospitals or clinics often lack an effective system to monitor whether patients are completing their home exercises between therapy sessions. This makes it difficult to accurately evaluate the child's progress or adjust the treatment plan based on the child's real performance at home.

As a result, a gap emerges between the therapy sessions conducted by specialists and the practical implementation of exercises at home, which may reduce the overall effectiveness of the treatment plan.

Therefore, this project aims to create a unified digital platform that provides an integrated monitoring system connecting specialists in hospitals or clinics with families. Through the platform, specialists can track the child's progress, review completed exercises, analyze performance, and update the therapy plan when needed. This helps improve the quality of follow-up and increases the effectiveness of the treatment process.

The platform can also be customized to match the branding and identity of each hospital or clinic that adopts the service, ensuring a unified system while maintaining each healthcare provider's visual identity.

Target Audience

The platform is designed to serve multiple user groups, including:

- Children with Sensory Processing Disorder challenges.
- Parents and family members responsible for supporting therapy exercises at home.
- Therapists and specialists who design therapy programs and monitor the child's progress.

Key Features

The proposed MVP will include several essential features:

- **Therapy Plan Management:** Specialists can create therapy plans and convert them into structured exercises and activities.
- **Home Exercise Guidance:** Parents receive clear instructions on how to perform sensory exercises with their child.
- **Progress Tracking:** Families can record completed activities and track the child's progress over time.
- **Specialist Monitoring:** Specialists can remotely review progress and adjust therapy plans when needed.
- **Communication Support:** The platform helps improve communication between families and specialists.

Expected Outcomes

The platform aims to improve the consistency of therapy exercises performed at home, strengthen communication between families and specialists, and provide better insights into the child's progress and development. By focusing on these core features, the MVP will provide a practical and impactful solution while remaining achievable within the project's development scope.

4. Idea Development Documentation

Overview of the Idea Development Process

In the early stages of the project, the team followed a structured approach to generate ideas, analyze them, and select the most suitable concept for implementation. This process involved several key stages, including research, brainstorming, idea evaluation, and final decision-making. Brainstorming sessions applied the "How Might We" questions technique to generate diverse ideas and examine problems from multiple perspectives.

Ideas Considered

During the brainstorming sessions, the team discussed four potential ideas before selecting the final project concept.

Idea 1 – SPD Support Platform

A digital platform designed to support children diagnosed with Sensory Processing Disorder (SPD) by providing a structured system of sensory exercises that can be performed at home. The platform enables specialists to convert therapy plans into clear exercises and activities that families can easily follow with their children.

The platform also provides a monitoring system for doctors and specialists in hospitals or clinics, allowing them to track patients' adherence to home exercises, review recorded progress, analyze performance, and modify the treatment plan when necessary.

Strengths

- Addresses a real problem faced by many families.
- Enhances communication between specialists and families.
- Provides a digital monitoring system that helps improve treatment effectiveness.
- Can be adopted by hospitals or clinics with customizable visual identity for each organization.

Weaknesses

- Requires accurate therapeutic content developed in collaboration with specialists.
- Needs a simple and user-friendly interface suitable for families.

Opportunities

- Growing demand for digital health solutions supporting children with special needs.
- Potential adoption by multiple hospitals and clinics, enabling platform scalability.
- Opportunity to expand into other therapy types beyond SPD in future versions.
- Alignment with government and NGO initiatives supporting children with disabilities.
- Limited direct competition in the market for this specific type of platform.

This idea was ultimately selected as the Minimum Viable Product (MVP) for the project.

Idea 2 – Virtual Penetration Testing Environment

A platform that provides a secure virtual environment where companies can test the security of their websites or applications by simulating penetration testing and identifying security vulnerabilities.

Strengths

- Related to the cybersecurity field.
- Useful for companies to test and improve their security systems.

Weaknesses

- Requires advanced technical infrastructure.
- High technical complexity for implementation.

Reason for Rejection

This idea was rejected due to its high technical complexity and infrastructure requirements, which exceed the scope and timeline of the project.

Idea 3 – Custom Dashboard Builder Platform

A platform that allows companies or organizations to create customized dashboards to manage their data and operational processes.

Strengths

- Useful for companies in managing data and operations.
- Potential to evolve into a commercial platform in the future.

Weaknesses

- Many existing competing solutions already exist in the market.
- Requires advanced customization features to stand out.

Reason for Rejection

This idea was rejected due to the strong competition in this field and the difficulty of offering a clear competitive advantage within the project scope.

Idea 4 – AI-Based Long Video Generation Tool

An artificial intelligence tool designed to generate long-form videos while maintaining coherence and context between different video segments.

Strengths

- An innovative idea based on AI technologies.
- Potentially useful for content creators.

Weaknesses

- Requires high computational resources.
- Technically complex to implement and requires significant infrastructure.

Reason for Rejection

The idea was rejected due to its high technical complexity and infrastructure requirements compared to the scope of the project.

Summary of the Selected Idea (MVP)

After evaluating all the proposed ideas, the team decided to select the SPD Support Platform as the project's Minimum Viable Product (MVP). The platform aims to help families perform sensory exercises at home in an organized way while providing a monitoring system that enables specialists in hospitals or clinics to track the child's progress remotely, review exercise performance, and modify the treatment plan when necessary.

Reason for Selecting the Idea

- It addresses a real problem faced by many families.
- It has a positive social impact in supporting children's development.
- It can be technically implemented within the project scope and available time.
- It offers future scalability to support multiple hospitals or clinics.

Expected Impact of the Project

The platform is expected to contribute to:

- Improving the monitoring of children's sensory exercises at home.
- Strengthening communication between specialists and families.
- Providing data that helps track the child's progress more effectively.
- Improving the effectiveness of therapy programs through continuous follow-up.

Conclusion

In conclusion, the team successfully completed Stage 1 by forming a collaborative team, exploring multiple project ideas, and selecting the SPD Support Platform as the MVP. This decision was based on its strong social impact, feasibility, and alignment with the team's technical capabilities. Moving forward, the team will focus on the design and development phase to bring this platform to life.